

HF Sinclair oil refinery in Anacortes (formerly Shell)

AIRPACT-6 Emissions Updates

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NW-AIRQUEST Fall Mtg
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DEPARTMENT OF ECOLOGY
State of Washington

AIRPACT-6 Point Source Update Process

Started with 2022 EMP Emissions

Emissions from facilities:

- 2022 Agency Submittals (Type A Majors)
- 2020 Agency Submittals (Type B Majors)
- 2020 - 2022 Toxics Release Inventory

Emissions from EPA:

- HAP-Augmentation (based on SCC)
- PM-Augmentation Gap-Fill (PRI, CON, FIL)
- PM Speciation (EC, OC, NO3, SO4, PMFINE)

AIRPACT-6 Point Source Update Process

States provided Add/Update/Remove list

Changes submitted by:
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Note that submitted emissions changes did not include PM speciation, PM augmentation, or HAP augmentation. These were inferred as needed.

2024 Emissions Adds/Updates:

Missing pollutants for existing processes used 2022/2024 ratio:

- PM Speciation from PM2.5-PRI
- PM Augmentation (PRI/FIL/CON)
- HAP Augmentation based on type
  - HAP Metals ratioed with PM2.5
  - HAP VOCs ratioed with VOC
  - Other HAPs pulled forward (no change)

All newly added process emissions used:

- default PM Speciation/Augmentation
- No HAP Augmentation

Files for AIRPACT-6

2022 EMP Initial Point Files

- EGU FF10 (all states)*
- Non-EGU FF10 (all states)*
- Oil & Gas FF10 (all states)
- Rail Yards FF10 (all states)
- Airports FF10 (all states)

AIRPACT-6 Point Files

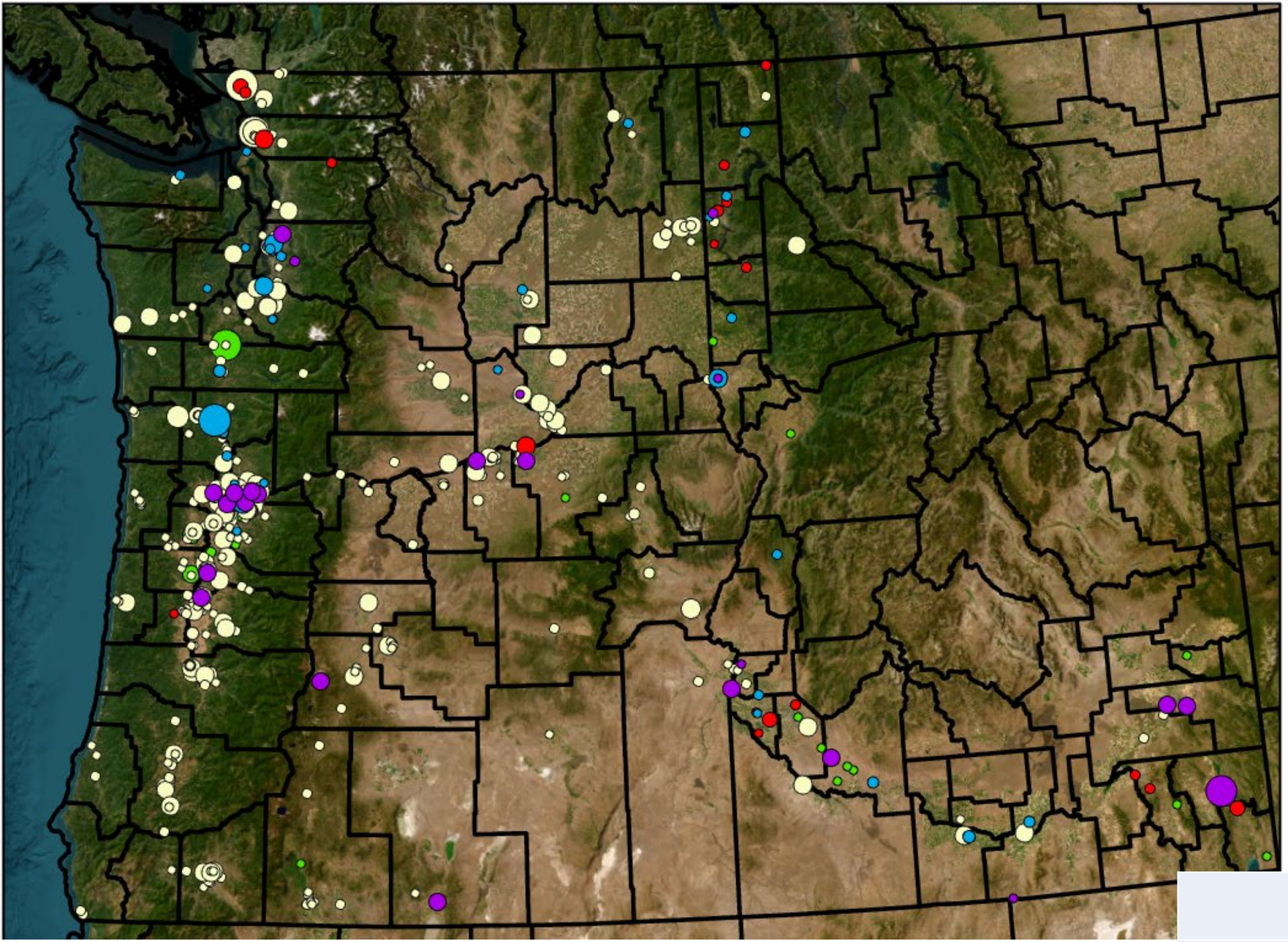
Updated Emissions
- Idaho FF10 (EGU/nonEGU/O&G)
- Oregon FF10 (EGU/nonEGU/O&G)
- Washington FF10 (EGU/nonEGU/O&G)

Other Point Files:
- Rail Yards FF10 (all states)
- Airports FF10 (all states)
- EGU FF10 (all other states)
- Non-EGU FF10 (all other states)
- Oil & Gas FF10 (all other states)

*The EMP splits facilities across EGU and nonEGU files based on SCC. However, this is not necessary to process emissions in SMOKE.

Point Sources with 100 ton +/- change in Q (NOx + VOC + PM10 + SO2 + NH3)

facility_id	State	FIPS	Name	Comment	dNOX	dVOC	dPM10	dSO2	dNH3
929311	Idaho	16083	AMALGAMATED SUGAR COMPANY - TWIN FALLS	Decrease	-45	-6	-19	-156	-51
2686011	Idaho	16001	ADA COUNTY LANDFILL - ACL & HHE	Increase	29	18	116	62	0
931611	Idaho	16027	AMALGAMATED SUGAR COMPANY - NAMPA	Increase	54	15	-3	1	57
2807311	Idaho	16027	PICKLES BUTTE SANITARY LANDFILL	Increase	6	0	99	19	0
948411	Idaho	16029	ITAFOS CONDA LLC	Increase	3	0	-1	119	0
15472711	Idaho	16055	AVISTA CORP	Increase	130	6	31	2	0
559811	Idaho	16077	J R SIMPLOT COMPANY-DON SIDING POCATELLO	Increase	68	2	49	875	198
23037411	Idaho	16019	FRAZIER INDUSTRIAL CO. - IDAHO FALLS	New Source	0	111	0	0	0
932011	Idaho	16029	RUVEON	New Source	1,977	3	128	397	0
23037211	Idaho	16055	RATHDRUM POWER LLC	New Source	57	4	11	3	27
23037011	Idaho	16075	CLAY PEAK MUNICIPAL SOLID WASTE LANDFILL	New Source	9	9	65	0	31
15472811	Idaho	16019	BUSCH AG RESOURCES INC - MALT PLANT	Shutdown	-65	-4	-118	-42	0
932011	Idaho	16029	P4 PRODUCTION LLC (TV FACILITY)	Shutdown	-1,787	-3	-325	-401	0
8355111	Idaho	16039	IDAHO POWER CO - EVANDER ANDREWS COMPLEX	Shutdown	-130	-6	-20	-2	0
2465111	Idaho	16049	IDAHO FOREST GROUP LLC - BENNETT-GRANGEVILLE	Shutdown	-105	-126	-25	-6	0
7321711	Idaho	16069	CLEARWATER PAPER CORP - PPD & CPD	Shutdown	-1,165	-288	-200	-44	-64
8417511	Oregon	41005	Willamette Falls Paper Company, Inc.	Shutdown	-201	-22	-4	-3	0
8057311	Oregon	41047	Covanta Marion, Inc.	Shutdown	-221	-2	-13	-3	0
8520811	Oregon	41051	Owens-Brockway Glass Container Inc.	Shutdown	-167	-1	-35	-73	0
4783711	Washington	53015	LANXESS Corporation	Decrease	-16	245	-1	-2	0
7000311	Washington	53015	Nippon Dynawave Packaging Co.	Decrease	45	-1	-12	-190	-4
4985311	Washington	53033	Ardagh Glass Inc.	Decrease	-129	-11	-23	-52	0
12626211	Washington	53041	Cardinal FG Company Winlock	Decrease	-457	7	25	21	0
12619111	Washington	53053	Waste Connections Inc Dba. Pierce Co Recycling Composting and Disposal LLC	Decrease	25	0	0	-131	0
8532811	Washington	53057	PSE FREDONIA	Increase	369	5	11	3	3
6212611	Washington	53073	PSE Ferndale Generating Station	Increase	78	16	18	7	7
6212111	Washington	53073	PSE WHITEHORN	Increase	448	2	10	3	0
17989711	Washington	53015	North Pacific Paper Corp. (NORPAC)	Increase	0	466	-7	0	0
7504511	Washington	53033	King Co Solid Waste Op Sec Cedar Hills	New Source	82	5	0	92	0
4880211	Washington	53027	COSMO Specialty Fibers, Inc.	Shutdown	-349	-26	-392	-269	0
6281311	Washington	53041	TransAlta Centralia Generation, LLC	Shutdown	-3,650	-96	-302	-1,072	0
4880911	Washington	53053	WestRock Tacoma Mill	Shutdown	-922	-107	-143	-171	-24



** Many "New Sources" are Type B landfills not in the 2022 EMP files*

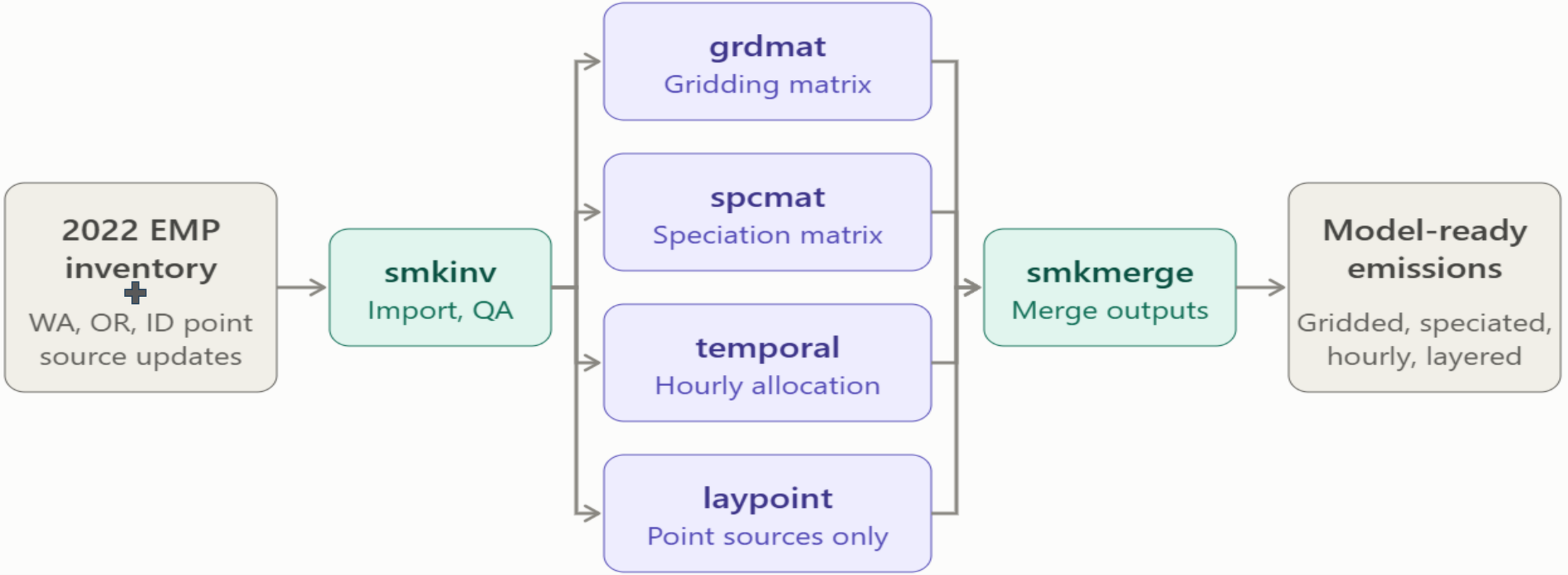
Point Emissions Changes

Total Annual Emissions

	CO	NOX	PM10	SO2	VOC	NH3
2022EMP	45,031	40,194	10,172	8,659	20,511	5,856
AP6_New	38,763	33,474	8,924	7,663	20,502	6,168

EI Update Status

- AP6 Point source replacement files delivered to WSU
  - Includes HAPs and PM sub-species
- WSU agreed to install all new files before end of year
  - Targeting 2027 as first full year of operational AIRPACT-6
- Next steps for EI updates
  - Sensitivity study to tune fugitive nonpoint dust emissions
  - Goal is to correct any domain-wide PM bias
    - Focus will be on clean/Spring days, when other PM sources are at a minimum and wet deposition is not a factor. April 2026 rerun?



Questions?

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